



Biodiesel Production in Tubular Microreactor: Optimization by Response Surface Methodology

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Abstract

Transesterification of crude cottonseed oil with methanol in the presence of catalyst (NaOH) in tubular microreactor has been investigated experimentally. The transesterification reaction was performed in a silicon tube of 0.8 mm inner diameter, mounted in serpentine manner configuration on an acrylic sheet. Influence of process variables such as reaction temperature (35–45 °C), NaOH concentration (0.5–1.5 wt%) and oil/methanol molar ratio (1:7–1:9) on fatty acid methyl ester (FAME) was studied. In order to further improve biodiesel yield, an experimental design was employed using the Box–Behnken method and analysis of variance. The %FAME was calculated by gas chromatography using methyl arachidate as an internal standard. Fourier transform infrared spectroscopy was used to investigate the functional groups present in biodiesel. Thermal stability of biodiesel was evaluated using a thermogravimetric analyser. The optimal condition found was oil/methanol molar ratio (1:8), catalyst concentration (1 wt%) and reaction temperature (45 °C) while %FAME yield of about 94.1% at a residence time of 90 s.

Keywords Tubular microreactor · Crude cottonseed oil · FAME · Response surface methodology

1 Introduction

Depletion of fossil fuel resources and several environmental issues have attracted the attention of many researchers in developing renewable and clean alternative fuels [1]. Due to this, more consideration has been paid to produce bio-fuels such as biodiesel which are renewable, biodegradable and non-toxic, with high flash point and good greenhouse emission profile [2,3]. Biodiesel is the free fatty acid methyl ester (FAME) derived from vegetable oils and animal fats [4]. When vegetable oil and animal fats reacts with alcohol (methanol/ethanol) in the presence of a catalyst (acid,

base, and enzymatic catalyst) alkyl ester that is biodiesel and glycerol as a spin-off are produced. Generally, selection of catalyst depends upon the free fatty acid (FFA) content of oil. Usually, a homogenous alkali catalyst, NaOH or KOH is preferred due to their operational simplicity, high catalytic activity and greater yield. NaOH is economically more favourable over KOH which is its key advantage. Furthermore, NaOH has low molecular weight compared to KOH it will have higher molar concentration even though weight concentration is equal and hence less amount of catalyst will be required. Widely preferred alcohol for the transesterification reaction is methanol owing to its low cost, physical and chemical advantages and its ability to quickly dissolve the alkali catalyst [5].

Although, numerous literature has revealed the importance of biodiesel, especially with respect to future energy needs and environmental issues, implementation of biodiesel still poses some problems [6,7]. Unavailability of sufficient cost-effective feedstock and large production time has increased the cost of biodiesel, which is a major obstacle to its commercialization.

Generally, biodiesel is prepared by various methods in batch reactors such as transesterification, pyrolysis and microemulsification [8]. However, these methods are very

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time-consuming and expensive. To overcome these limitations, many researchers are moving towards continuous processes [9,10] which have many advantages over batch reactor like less residence time, high surface-to-volume ratio, higher heat and mass transfer rates, ease in controlling process parameters during operation and overall safety of the process is significantly enhanced [11–14]. In petrochemical industries due to low profit margins, strong competition and high safety regulation, there is a need for continuous process which can be cost-effective, safe and efficient. Microreactor is a device in which chemical reactions take place in a confinement with inner diameter below 1 mm, the most common type of such confinement is a microchannel [11]. As the transesterification reaction is mass transfer limited, the microreactor can serve to increase the reaction rate and the product synthesis per reactor volume by reducing the time of production from h to min [5,15]. Canter et al. reported that operating temperature was the negligible parameter effecting the biodiesel yield however, the inner diameter of microreactor had a virile control on transesterification reaction [16]. In addition, Jovanovic et al. patented that the both effect of channel thickness (100 μm) and residence time (10 min) had a strong influence on conversion of soyabean oil to biodiesel (91% conversion) [17].

The feedstock for biodiesel production is influenced mainly by climate and local soil conditions; hence, different countries have focused on distinctive raw materials to produce biodiesel. The USA uses soyabean oil as a raw material, whereas Germany prefers rapeseed oil. India has considered jatropha as an appropriate feedstock for biodiesel production. But the higher gestation period of the biodiesel crop (3–5 years) results in a longer payback period for farmer which is the major obstacle for large-scale cultivation of jatropha. Looking for an alternative, India is the second biggest cultivator of cotton in the world and cotton plant stand out to be the most vital raw materials for the textile industry. Cotton plant yields 800–950 kg seeds per hectare and around 140 kg oil per hectare [18]. Earlier, cottonseed oil was used in the food industry, but due to the imbalance, percentage of omega-3 and omega-6 fatty acid in cottonseed oil is regarded as harmful for consumption so its demand in the food industry is diminishing [19]. Hence, cottonseed oil has great potential of becoming an inexpensive raw material to produce biodiesel in India.

There are several process variables that influence the yield of biodiesel prepared using crude cottonseed oil in microreactor. For the optimization of these process variables, design of experiment (DOE) which is one of the most prominent statistical tools has been employed. Response surface methodology (RSM), which comprises of mathematical and statistical techniques, is valuable for studying the interactive effect of process variables. Due to fewer exploratory runs and less time, Box–Behnken factorial design is one of the fre-

quently preferred design among the various RSM designs, which provides more efficient statistical optimization [20–22]. Here, the influence of various independent variables on %FAME (dependent variable) was examined by Box–Behnken design.

In the present study, tubular microreactor of silicon tube (0.8 mm ID) is used as a continuous flow reactor for biodiesel production from crude cottonseed oil using NaOH as a catalyst and Box–Behnken design as a statistical tool which is not reported elsewhere in the literature to best of our knowledge. The impact of process variables such as NaOH concentration, molar ratio of methanol to cottonseed oil and reaction temperature on the transesterification of triglycerides was examined. In addition, optimal reaction conditions for producing the maximum methyl ester yield and the relation between the operating parameters were determined using Box–Behnken experimental design method.

2 Materials and Methods

2.1 Materials

Crude cottonseed oil was supplied by ginning and pressing as feed for transesterification reaction. Methanol (99.5%) and sodium hydroxide pellets were procured from Merck, India. Methyl arachidate (99.99%) was obtained from Sigma-Aldrich as internal standard for GC analysis. All other chemicals were analytical grade and used as received.

2.2 Experimental Set-up and Procedure

A schematic representation of the experimental set-up for biodiesel production is depicted in Fig. 1. Tubular microreactor consists of a T-shaped microfluidic junction and silicon microtube with an internal diameter of 0.8 mm mounted on acrylic sheet in a serpentine manner [23]. The microreactor was submerged in a water bath to obtain the desired temperature. At the first step, in order to prepare the methanol/sodium hydroxide solution, it was necessary to calculate the amount of catalyst. The methanol/sodium hydroxide solution and oil were fed separately in the syringes and driven by syringe pump into two input channels of the T-micromixer. Both feeds were mixed well and passed through the silicon tube. The flow rates of the two feed streams were adjusted accordingly to maintain molar ratio of cottonseed oil to methanol (1:7–1:9). The resultant product was collected from the exit end of microreactor. The outlet sample was poured in a separating funnel to split the two phases, FAME (golden colour) at the top and glycerol (dark brown colour) at the base. Finally, FAME phase was centrifuged for 5 min at 10,000 rpm to remove residual catalyst and glycerol.

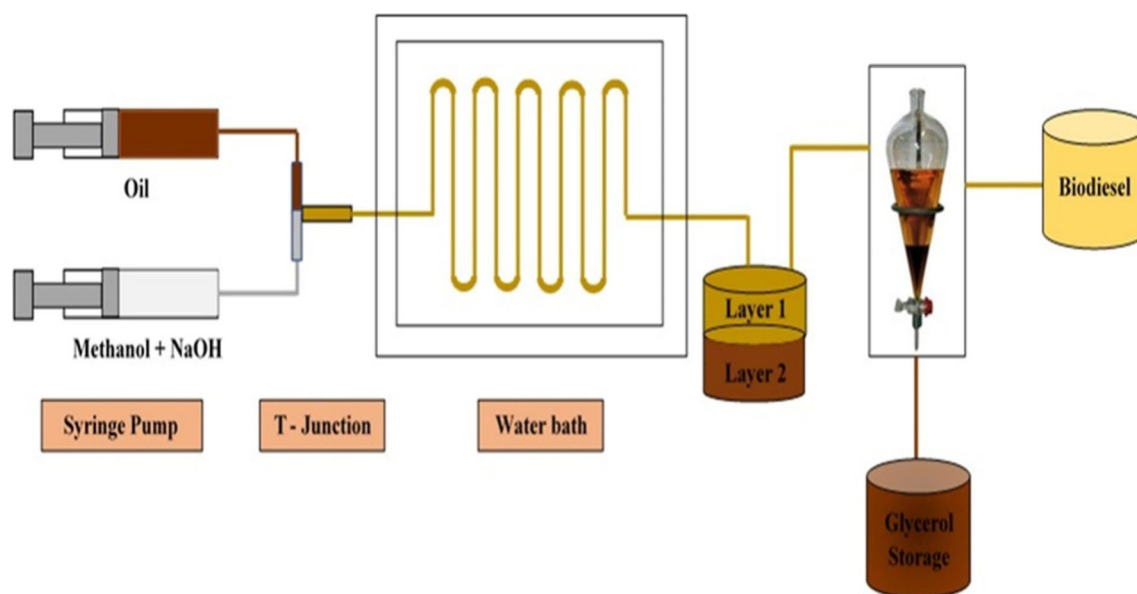


Fig. 1 Experimental set-up for production of biodiesel in tubular microreactor

2.3 Characterization of Biodiesel

2.3.1 Gas Chromatography Analysis

The produced biodiesel was examined by Agilent 7890 series GC equipped with flame ionization detector. HP-INNO wax column (30m, 0.25 mm, 0.25 μ m) was used in this study [24]. Helium was introduced in column at 1 mL/min constant flow condition. Inlet column temperature was 60 °C for 2 min withhold at 240 °C for 7 min. The temperature of flame ionization detector was set at 250 °C. Methyl arachidate (retention time 22.3750 min) was utilized as a reference for the GC. The 2 μ L sample of produced biodiesel and internal standard was injected into the GC. The total %FAME content was evaluated using the integrated peak area of all the FAMEs recognized in the sample. The analysis of FAME wt% for the prepared biodiesel sample was calculated by method of Wang et al. [25] using following Equation:

$$\%FAME = \frac{\text{Area of all FAME}}{\text{Area of all reference}} \times \frac{\text{Weight of reference}}{\text{Weight of biodiesel sample}} \times 100 \quad (1)$$

2.3.2 Fourier Transform Infrared (FTIR) Spectroscopy

To investigate the functional group present in biodiesel and to confirm the formation of ester group, infrared spectra study of biodiesel was performed using fourier transform infrared spectroscopy (FTIR-8400; Shimadzu, Asia pacific Pvt. Ltd. Singapore).

2.3.3 Thermogravimetric Analysis

The thermogram of biodiesel was examined by thermogravimetric analyser (TGA-50, Shimadzu, Japan) using aluminium pans. The thermal analysis was conducted at a heating rate of 10 °C/min under highly purified nitrogen at flow rate of 50 mL/min from 10 to 600 °C.

2.4 Design of Experiments and Optimization

To evaluate the effect of independent variable on %FAME, Box–Behnken factorial design (three-factor and three-level) was employed. In this work, Box–Behnken factorial design is used to understand the effect of each variable on the continuous transesterification of cotton seed oil. The aim of selecting Box–Behnken design is to discover the optimum conditions and to achieve the highest possible biodiesel yield with minimum number of experimental runs. Selection of levels for each factor was based on the trial experiments performed to investigate the effect of process parameters on transesterification reaction of crude cottonseed oil in microreactor. The independent variables selected were the reaction temperature (X_1 °C), catalyst concentration (X_2 wt%) and the molar ratio of methanol to cottonseed oil (X_3), which were varied at three levels (−1, 0, +1). The %FAME was selected as dependent variables. Analysis of statistical experimental data was performed using design-expert® Software (Stat-Ease Inc., Minneapolis). The framework of design is shown in Table 1.



Table 1 Experimental range and levels of variable

Independent variable	Low level (−1)	Medium level (0)	High level (+1)
X_1 = Temperature (°C)	35	45	55
X_2 = Catalyst (wt%)	0.5	1	1.5
X_3 = Molar ratio	1:7	1:8	1:9
Dependent variable			
Y_1 = FAME%			

Effects of independent variables upon response were modelled using the following quadratic model [21–23].

$$Y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_{12}x_1x_2 + b_{13}x_1x_3 + b_{23}x_2x_3 + b_{11}x_1^2 + b_{22}x_2^2 + b_{33}x_3^2 \quad (2)$$

where Y is the response, b_0 is the intercept and b_1 , b_2 , b_3 , b_{12} , b_{13} , b_{23} , b_{11} , b_{22} and b_{33} are the regression coefficient. x_1 , x_2 , and x_3 are individual effect; x_1x_2 , x_1x_3 , and x_2x_3 are interaction effect; and x_1^2 , x_2^2 and x_3^2 are the quadratic effect. The significance of statistical parameters and model was evaluated at 95% confidence level ($P < 0.05$) [26].

3 Results and Discussion

Transesterification of soyabean oil with methanol and KOH in a microreactor has been successfully investigated by Rahimi et al. [27] and the reaction was performed at specific condition of 0.8 mm hydraulic diameter in circular tube. The optimum condition was found at temperature (60 °C), molar ratio (9:1) and KOH concentration (1.2 wt%) with %FAME of about 89%. In the present study, the optimum condition was found to be at 45 °C reaction temperature, molar ratio methanol to oil (8:1) and 1 wt% catalyst concentration with %FAME yield of 94.1%.

Zhang et al. [28] reported that the biodiesel could be produced using crude rapeseed oil and cottonseed oil in microreactor. About 95% biodiesel yield might be achieved using capillary microreactor of 0.25 or 0.53 mm ID, methanol/oil molar ratio of 6 and 1% of KOH concentration at 60 °C temperature. However, 94.1% of biodiesel yield was achieved in residence time of 90 s in this study.

Rahimi et al. [29] successfully prepared biodiesel using the ultrasound-assisted soyabean oil methanolysis with KOH as a catalyst at different reaction conditions in a microreactor. 97.1% methyl ester yield was obtained at optimal conditions of KOH concentration of 1.29%, adjusting the methanol/oil ratio of 6:1 and 47 °C of reaction temperature. In this research, by using only tubular microreactor, 94.1 FAME% was achieved.

3.1 Physicochemical Properties

The result of various physicochemical properties of biodiesel and crude cottonseed oil along with the standards (ASTM D6751-02) is shown in Table 2 [30]. Most of the properties are in good agreement with the standards which indicates that crude cottonseed oil is an excellent source of raw material for biodiesel production.

3.2 Statistical Analysis of %FAME

Box–Behnken factorial design was applied to correlate experimental process variables and the transesterification efficiency. Reaction temperature (X_1), NaOH concentration (X_2) and molar ratio (X_3) were selected as independent variables, while %FAME (Y_1) was dependent variable. The %FAME for different experimental runs was found in the range of 79.2–94.1% listed in Table 3.

Diagnostics case statistics for (%FAME) are given in Table 4. Residual value (difference between predicted response and actual response) was very less for each run, which signifies the model was best fitted. The quadratic model in terms of the coded values of process variables and %FAME (Y_1) as response became,

$$Y_1 = 93.44 + 1.00X_1 + 1.93X_2 + 2.15X_3 - 1.00X_1X_2 + 0.10X_1X_3 + 0.85X_2X_3 - 3.52X_1^2 - 3.72X_2^2 - 6.77X_3^2 \quad (3)$$

The above quadratic equation for a response indicates the positive and negative value of the factors which represent the synergistic and antagonistic effect. In case of response Y_1 , synergistic effect represents an effect that favours the response, while the antagonistic effect represents the inverse relationship [31].

The optimized value of correlation coefficient (0.9906) for the quadratic model of response Y_1 representing the good fit is illustrated in Table 5.

The significance of the regression coefficients was evaluated through p values less than 0.05. The p (< 0.0001) value of the response Y_1 was significant at $p < 0.05$ (F -value = 81.77), listed in Table 6. The model terms X_1 , X_2 , X_3 , X_1X_2 , X_1^2 , X_2^2 , and X_3^2 were statistically significant, whereas X_1X_3 and X_2X_3 interaction terms were insignificant and removed

Table 2 Characterization of crude cottonseed oil, biodiesel, and ASTM standard

Properties	Unit	Crude cottonseed oil	Biodiesel from cottonseed oil	ASTM D6751 standard
High heating value	MJ/kJ	43.28	42.44	–
Acid value	mg/KOH	0.42	0.28	0.50 max
FFA	%	0.21	0.14	–
Specific gravity	–	0.9624	0.9306	–
Saponification value	mg/g	173.91	123.42	–
Ester value	mg/g	173.49	123.14	–
Odour	–	Agreeable	Agreeable	–
Colour	–	Brown	Golden	–
Moisture content	wt%	0.027	0.027	0.050 max
Flash point	(°C)	238	167	100–170
Fire point	(°C)	–	180	–
Cloud point	(°C)	–3.7	6.5	–3 to 12
Pour point	(°C)	–3	4.6	–15 to 10
Aniline point	(°F)	–	143.6	–
API gravity	–	–	34.97	–
Diesel index	–	–	50.22	50
Kinematic viscosity	mm ² /s	32.33	7.07	1.9–6.0

Table 3 Experimental design and results

Experimental run	Temperature (°C)	Catalyst (wt%)	Molar ratio	%FAME
1	45.00	1.50	9.00	88.4
2	45.00	1.50	7.00	81.7
3	45.00	1.00	8.00	94.1
4	45.00	1.00	8.00	92.5
5	55.00	1.00	9.00	86.4
6	35.00	1.00	7.00	80.1
7	45.00	1.00	8.00	93.1
8	45.00	0.50	9.00	82.5
9	35.00	1.00	9.00	83.5
10	45.00	1.00	8.00	94.0
11	35.00	1.50	8.00	88.3
12	55.00	1.00	7.00	82.6
13	55.00	0.50	8.00	86.1
14	55.00	1.50	8.00	87.6
15	35.00	0.50	8.00	82.8
16	45.00	0.50	7.00	79.2
17	45.00	1.00	8.00	93.5

from the regression model. The adequate precision value of 23.784 (signal-to-noise ratio greater than 4 is desirable) signifies an adequate signal. The significance and adequacy of the established model were further elaborated by the high value of adjusted and predicted *R*-squared (0.9785 and 0.9095, respectively). Both the values of adjusted and predicted *R*-squared were in reasonable agreement (suggested difference is less than 2), which signifies the quadratic model predicts the response well [32].

3.3 Response Surface Estimation

The interactive influence of the process parameters such as reaction temperature, catalyst concentration and oil/methanol molar ratio on %FAME was examined by plotting 3-dimensional surface curves across the any two independent variables. The surface plot between catalyst concentration and reaction temperature against %FAME is shown in Fig. 2. It was observed that the %FAME increases from –1 to 0



Table 4 Diagnostic case statistics for various response variables

Experimental run	Response variables	Actual value	Predicted value	Residual value
1	Y_1	88.4	87.88	0.52
2	Y_1	81.7	81.88	−0.18
3	Y_1	94.1	93.44	0.66
4	Y_1	92.5	93.44	−0.94
5	Y_1	86.4	86.4	0.00
6	Y_1	80.1	80.10	0.00
7	Y_1	93.1	93.44	−0.34
8	Y_1	82.5	82.33	0.17
9	Y_1	83.5	84.20	−0.70
10	Y_1	94.0	93.44	0.56
11	Y_1	88.3	88.13	0.17
12	Y_1	82.6	81.90	0.70
13	Y_1	86.1	86.27	−0.17
14	Y_1	87.6	88.12	−0.52
15	Y_1	82.8	82.27	0.53
16	Y_1	79.2	79.72	−0.52
17	Y_1	93.5	93.44	0.060

Table 5 Summary of regression analysis for response Y_1

Models	R^2	Adjusted R^2	Predicted R^2	SD	Press	Remarks
Response Y_1						
Linear	0.1777	−0.0121	−0.2182	5.15	511.56	Suggested
2FI	0.1942	−0.2893	−1.0485	5.82	860.27	
Quadratic	0.9906	0.9785	0.9095	0.75	38.02	
Cubic	0.9958	0.9833		0.66		

Table 6 ANOVA for the fitted quadratic model

Source	DF	Sum of squares	Mean square	F value	p value
Model for Y_1	9	415.99	46.22	81.77	< 0.0001
X_1	1	8.00	8.00	14.15	0.0071
X_2	1	29.65	29.65	52.44	0.0002
X_3	1	36.98	36.98	65.42	< 0.0001
X_1^2	1	52.17	52.17	92.29	< 0.0001
X_2^2	1	58.27	58.27	103.08	< 0.0001
X_3^2	1	192.98	192.98	341.39	< 0.0001
X_1X_2	1	4.00	4.00	7.08	0.0325
X_1X_3	1	0.040	0.040	0.071	0.7979
X_2X_3	1	2.89	2.89	5.11	0.0582

levels of catalyst and reaction temperature, but with further increase in levels (0 to +1) of catalyst and reaction temperature the %FAME decreases. The decrease in %FAME might be because of evaporation of methanol, as temperature reaches to its boiling point. It may also be due to increase in temperature has high tendency of promoting the saponification process by the base catalyst [33].

The effect between catalyst concentration and molar ratio on the %FAME is displayed by the response surface 3D contour plot shown in Fig. 3. The elliptical shape of the contour plot represents a good interaction between the two variables, whereas circular shape of the plot indicates no interaction of the variable. From the curve, it is clearly seen that increase in oil/methanol molar ratio is equally demanding an increase in

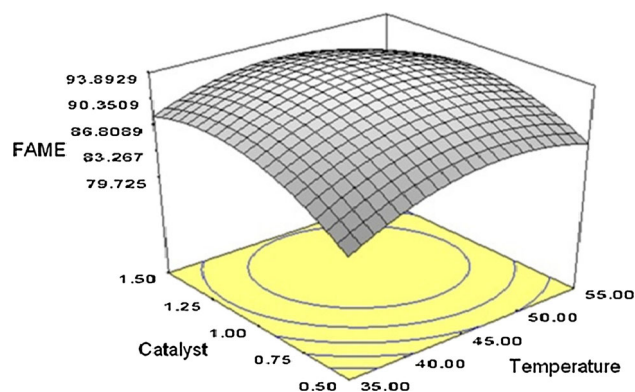


Fig. 2 Effect of catalyst concentration (wt%) and reaction temperature ($^{\circ}\text{C}$) against %FAME

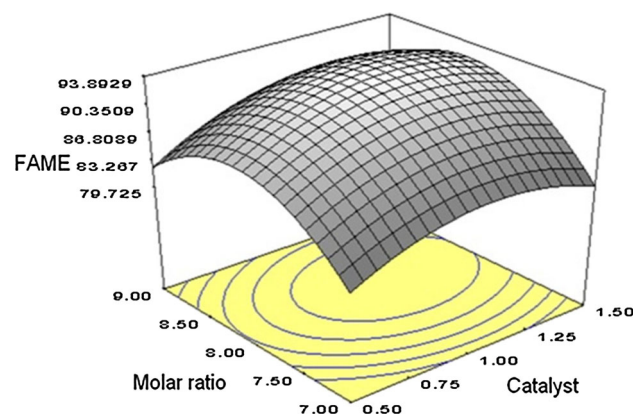


Fig. 3 Effect of molar ratio and catalyst concentration (wt%) on %FAME

catalyst amount. The %FAME increases from -1 to 0 levels of molar ratio and catalyst, but with additional increase in levels %FAME decreases. As increment in concentration of catalyst from -1 to 0 levels increases the active sites available, it would help to increase the reaction rate and give better conversion. Further increase in the catalyst level (0 to $+1$) it would participate in saponification process which reacts with triglyceride to form soap and water and causes problem on biodiesel yield [34].

The influence of oil/methanol and reaction temperature on %FAME is illustrated in Fig. 4. It is interpreted that there is a positive influence of rapid increase in temperature and oil/methanol molar ratio of %FAME as both the factors have similar goals of enhancing the rate of reaction. The %FAME increases from lower level to higher levels (-1 to 0) of molar ratio and reaction temperature, but additional increase in levels of molar ratio and reaction temperature shows inverse effect. At higher molar ratio, the unreacted methanol acts as a stabilizer and makes separation process of glycerol and methyl ester complicated which might be the reason for decrement in %FAME [27].

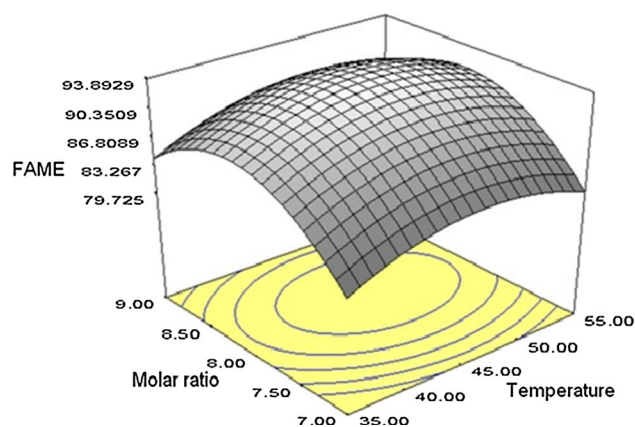


Fig. 4 Effect of molar ratio and reaction temperature ($^{\circ}\text{C}$) on %FAME

The coordinates of the stationary point (where the slope of the contour is zero) within the highest contour levels in each plot would correspond to the optimal conditions. The optimum values of the variables are 45°C reaction temperature, 1 wt% catalyst concentration and $1:8$ oil/methanol molar ratio. The predicted value at these conditions was 93.44% , whereas the experimental response value was found to be 94.1% . It signifies that the suggested quadratic model correctly explains the interaction of process variables on %FAME.

3.4 Fourier Transform Infrared (FTIR) spectroscopy

The FTIR spectra of biodiesel prepared from crude cottonseed oil at temperature 45°C , NaOH concentration 1 wt% and oil/methanol ratio $1:8$ are shown in Fig. 5. From the results, the sharp peak at 1741.65 cm^{-1} corresponds to (C–O medium stretch/ester) confirms the formation of biodiesel. The discernible peaks of note were recorded at 721.79 cm^{-1} (C–H weak rocking/alkanes), 846.71 cm^{-1} (C–C strong bend/aromatics), 1034.22 – 1195.05 cm^{-1} (C–C/C–O weak stretch/alcohols), 1361.48 cm^{-1} (O–H weak double bending/alkynes), 2853.48 cm^{-1} (C–H weak stretch) and 3009.15 – 3369.07 cm^{-1} (O–H strong stretch) [35].

3.5 Thermogravimetric Analysis (TGA)

The thermal stability of prepared biodiesel at optimum reaction condition was investigated by TGA. The TGA curve involves a weight loss as the oil was heated in the temperature range of 10 – 600°C . The temperature scans of biodiesel in nitrogen atmospheric heating are shown in Fig. 6.

The figure shows the decomposition and weight loss of the biodiesel sample with respect to temperature. The weight loss occurs because of evaporation and/or combustion of methyl ester present in biodiesel [36]. The thermogram of



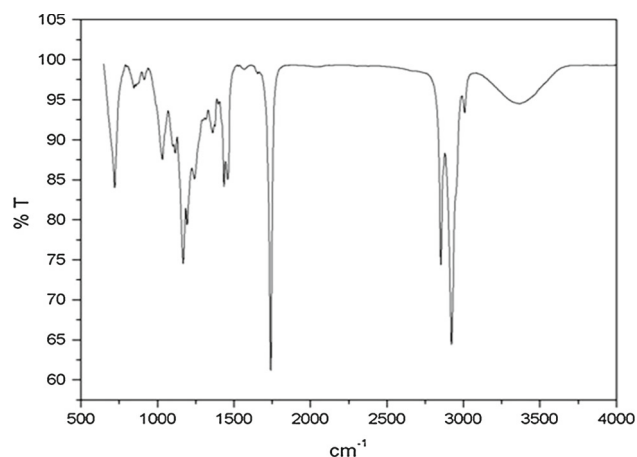


Fig. 5 FTIR spectra of prepared biodiesel

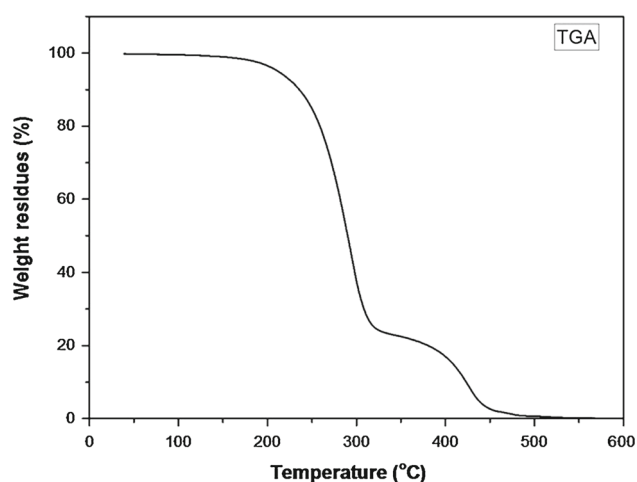


Fig. 6 TGA thermogram of biodiesel from crude cottonseed oil in nitrogen atmosphere

biodiesel presented two steps thermal decomposition. In inert atmosphere, the presented two steps are ascribed to the vapourization and/or pyrolysis of the methyl ester, generally methyl arachidate which is the main component of prepared biodiesel [37]. Almost negligible weight losses are analysed in the temperature range of 1–200 °C which indicates that methanol and moisture content in biodiesel is minimal. A rapid change was observed at temperature 210 °C and 99.854% weight loss occurred at 550 °C which represents complete decomposition of methyl esters present in biodiesel. High onset temperature (T_{on}) at 252.08 °C indicates high thermal stability.

4 Conclusion

We have successfully prepared biodiesel from crude cottonseed oil in a continuous mode of tubular microreactor. Optimization study was performed by Box–Behnken method

of experimental design. Biodiesel with the maximum percentage of methyl ester, 94.1% was produced at oil/methanol molar ratio 1:8, catalyst concentration 1 wt% NaOH and reaction temperature of 45 °C with a residence time of 90 s. The analysis of variance (ANOVA) results revealed adequacy of the model. TGA analysis showed thermal stability of biodiesel, whereas functional groups present in biodiesel was confirmed by FTIR spectra. Even though microreactor technology possesses numerous advantages, a major challenge remains its scale-up. Many of the substantial advantages of these reactors are related to small dimensions. Therefore, a new set of design consideration is necessary which does not impact these characteristics and provides required volume for industrial application. Parallel numbering up by using several reactors in parallel is one of the most common strategy in scale-up of microreactor. By this method, hydrodynamics and transfer characteristics can be retained but complex fluid flow distribution and control is a challenge. Additionally, handling of solids in these tubular reactors can cause clogging which is also a hurdle. Hence, more efforts are needed in fabricating and scale-up of this technology.

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